Assignment 3: Implementing and Analyzing TD Algorithms

Due: Tuesday June 2, before midnight

Objective:

To gain hands-on experience in implementing SARSA and Q-learning algorithms in gridworld.

Tasks:

I. Implement SARSA algorithm: **(10 points)**
II. Implement Q-learning algorithm: **(10 points)**
III. Analyze SARSA and Q-learning algorithms. (**40 points**)
    a. Plot the average reward as a function of the number of steps (both algorithms)
    b. 2. Generate the above plot for different values of epsilon (epsilon = 0.1, 0.5, 1) (both algorithms).
    c. 3. How different values of epsilon affect the training in both algorithms? (both algorithms)
    d. 4. What is the optimal value for epsilon? (both algorithms)

Submission:

Python Code: Fully documented code implementing the algorithms. **(The points divided in the tasks total of 60 points)**

Report: A comprehensive report detailing the algorithm's results, and an analysis describing task III. **(40 points)**

The Problem Description:

The code snippet is provided, and you can check the details of the grid world implementation and the environment dynamics there.

States: the different position on the grid. S indicates the start state and G indicates the final state.

Actions: up (0), right (1), down (2), and left (3).

Rewards: -5 for bumping into a wall, +10 for reaching the goal, and 0 otherwise.

Discount factor: γ=0.9

Note: You are given a code snippet. Feel free to use it or use your own implementation.